

**Feature:**

- S35353RGBSK6812MINID001 is a IC controlled RGB led source, led package size is 3535.
- This product doesn't contain restriction Substance, comply ROHS standard
- Wide viewing angle 120 deg.
- The output is driven by patented PWM technology, which highly performance on color consistency of the pixels.

Description:

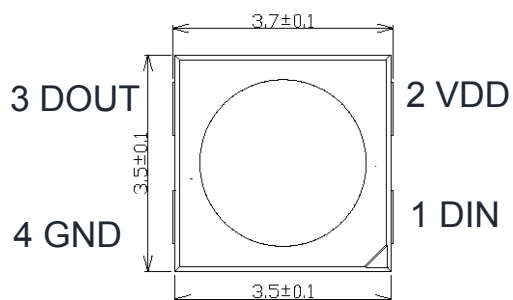
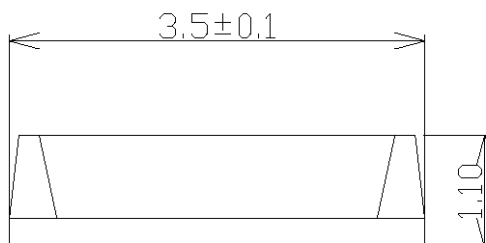
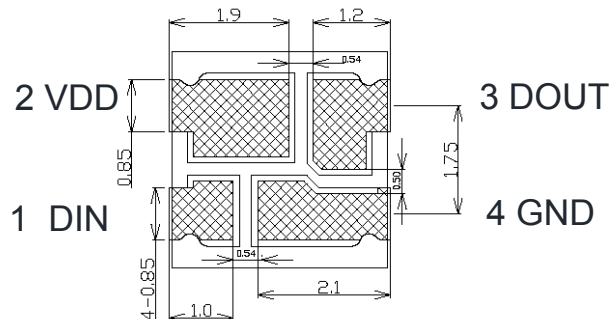
- Form a complete control of pixel, color mixing uniformity and consistency
- The built-in power on reset and reset circuit, the power does not work
- Gray level adjusting circuit (256 level gray scale adjustable)
- Using a typical data transmission frequency of 800 Kbps, when the refresh rate of 30 frames per sec

Application:

- LED full color module
- LED pixel screen
- Vehicle Audio/Stereo

Certification Compliance:

- ISO9001
- RoHS Compliant
- MSL:5a

**Dimension:****BOTTOM VIEW****Notes:**

1. All dimensions are in millimeters .
2. All dimensions tolerance is ± 0.1 mm unless otherwise noted.

**Optical Characteristic (T=25 °C)**

Color	S35353RGBSK6812MINID001 12mA		
	Dominate Wavelength (nm)	Luminance (mcd)	luminous flux(lm)
Red	620-625	240-450	
Green	520-530	815-1275	
Blue	460-470	160-320	

Absolute Maximum Rating

Parameter	Symbol	Rating	Unit
Power supply voltage	VDD	+3.7~+5.5	V
Logic input voltage	VIN	-0.5~VDD+0.5	V
Operation Temperature	Topr	-40 ~ 85	°C
Storage Temperature	Tstg	-40 ~ 85	°C
Reflow Peak Temperature	Tsol	< 10S @240°C	°C
ESD pressure(HBM)	VESD	2K	V
ESD pressure(DM)	VESD	200	V

1. Red chip Tj must be kept below 100°C; Green & blue chip Tj must be kept below 110°C.



Electrical Parameters

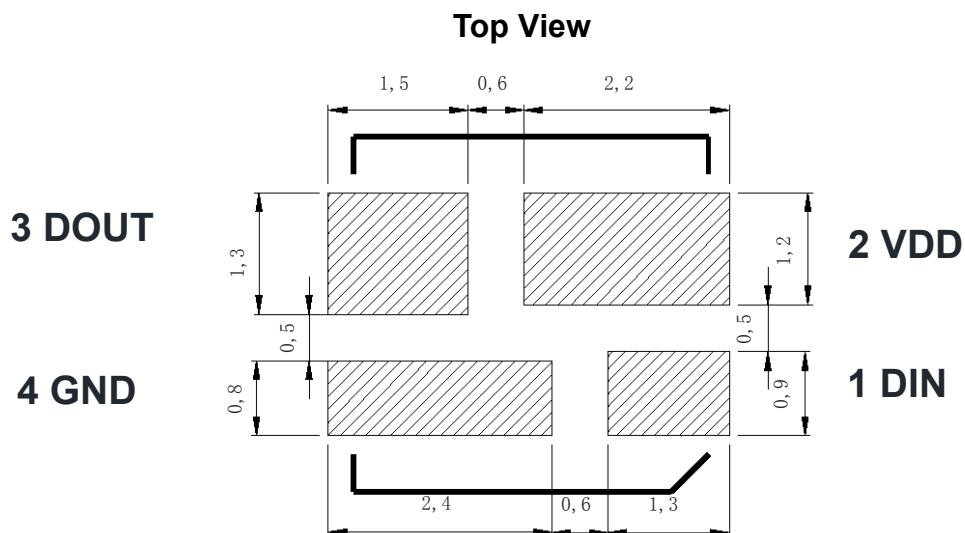
(unless otherwise specified, TA=-20 ~ +70 °C, VDD=4.5 ~ 5.5V, VSS=0V)

Parmeter	Symbol	Min	Typ.	Max	Unit	Test conditions
The chip supply voltage	VDD	---	5.2	---	V	---
The signal input flip threshold	VIH	0.7*VDD	---	---	V	VDD=5.0V
	VIL	---	---	0.3*VDD	V	
The frequency of PWM	FPWM	---	1.2	---	KHZ	---
Static power consumption	IDD	---	0.65	---	mA	---

PIN configuration

NO	Symbol	Function description
1	DIN	Control data signal input
2	VDD	Power supply LED
3	DOUT	Control data signal output
4	GND	Ground

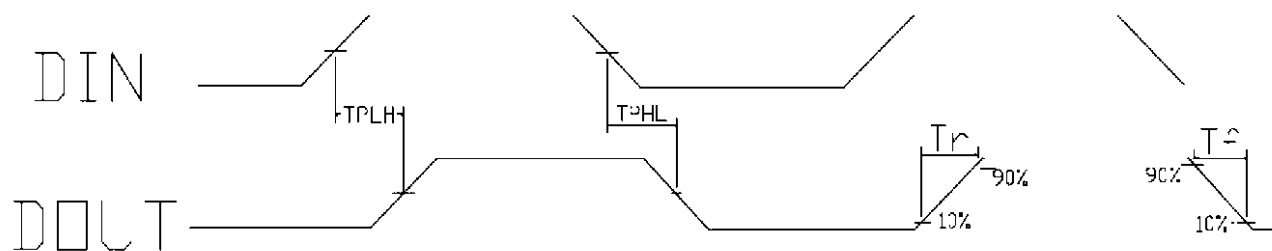
Recommended Dimensions for PCB products





Switching characteristics (Ta=25 °C)

Parameter	Symbol	Min	Typical	Max	Unit	Test conditions
The speed of data transmission	fDIN	---	800	---	KHZ	The duty ratio of 67% (data 1)
DOUT transmission delay	TPLH	---	---	500	ns	DIN→DOUT
	TPHL	---	---	500	ns	
IOUT Rise/Drop Time	Tr	---	100	---	ns	VDS=1.5 IOUT=13mA
	Tf	---	100	---	ns	



The data transmission time

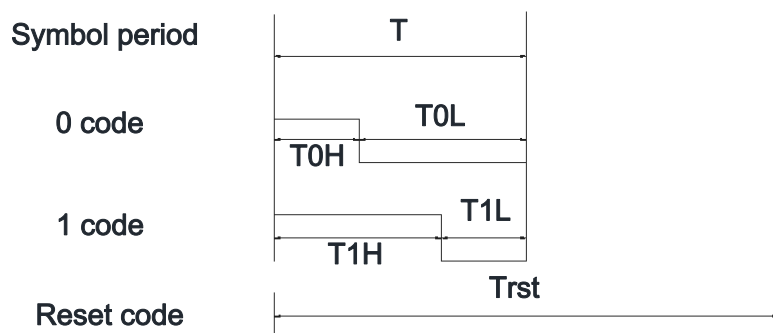
Name		Min.	Standard value	Max.	Unit
T	Code period	1.20	--	--	μs
T0H	0 code, high level time	0.2	0.32	0.4	μs
T0L	0 code, low level time	0.8	--	--	μs
T1H	1 code, high level time	0.58	0.64	1.0	μs
T1L	1 code, low level time	0.2	--	--	μs
Trst	Reset code, low level time	>80	--	--	μs

1. The protocol adopt unipolar zeroing code, Each symbol must have a low electrical level, each start protocol has high electrical level, average time of high electrical level define "0" or "1".
2. When programming, single code time must be equal or above 1.2μs
3. High electrical level time of code "1" & "0" should apply to upmentioned table, low electrical level time of code "1" & "0" should less than 2μs.

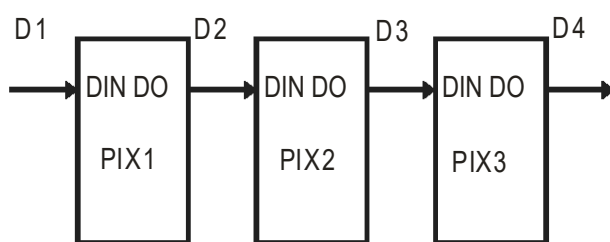


Timing waveform

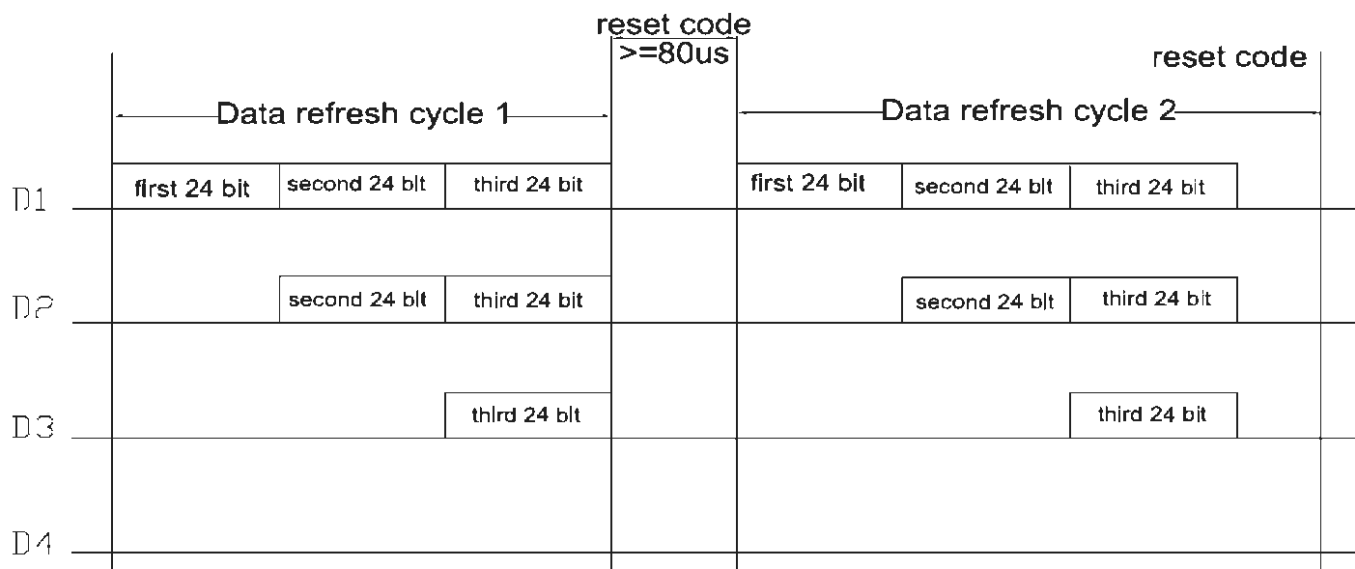
Input code:



Connection mode:



The method of data transmission



Note: MCU send data to D1; D2, D3, D4's data from cascade circuit after automatic shaping.

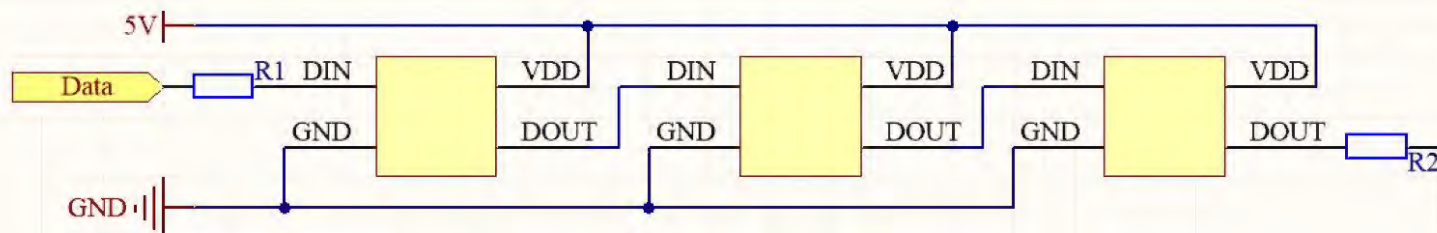


The data structure of 24bit

G7	G6	G5	G4	G3	G2	G1	G0	R7	R6	R5	R4
R3	R2	R1	R0	B7	B6	B5	B4	B3	B2	B1	B0

Note: high starting, in order to send data (G7 - G6 -B0)

The typical application circuit



In practice, in order to protect IC from damage which caused by surge during power on/off process, resistor should be fixed with input/output lead of IC; Meanwhile, decoupling capacitor is needed between leds, it will help IC co-work better.

Application: for soft led strip and hard led strip, leds close each other, recommend to fix resistors in circuit of input and output end of signal & clock line, resistor $R1=R2$ about 500 ohms.

Application: for module or Alien appearance products, leds have far transmission distance, based on different wire and transmission distance, you should choose right resistor depend on actual situation.



Characteristic Curves:

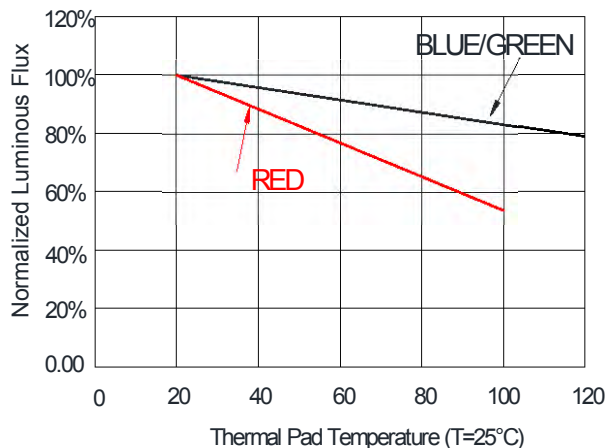


Fig.1 Thermal Pad Temperature vs. Relative Light Output

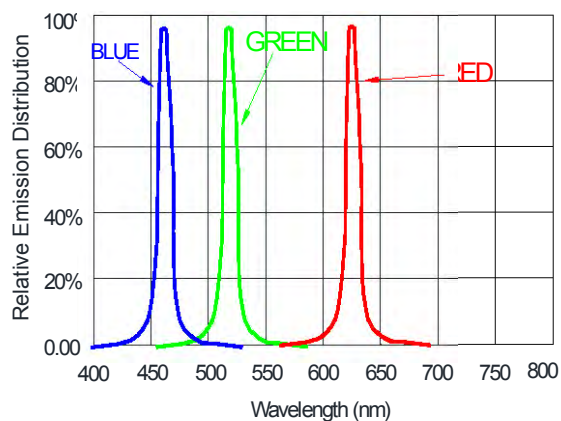


Fig.2 Wavelength Characteristics

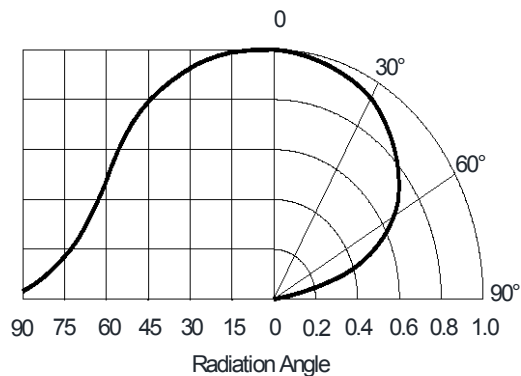


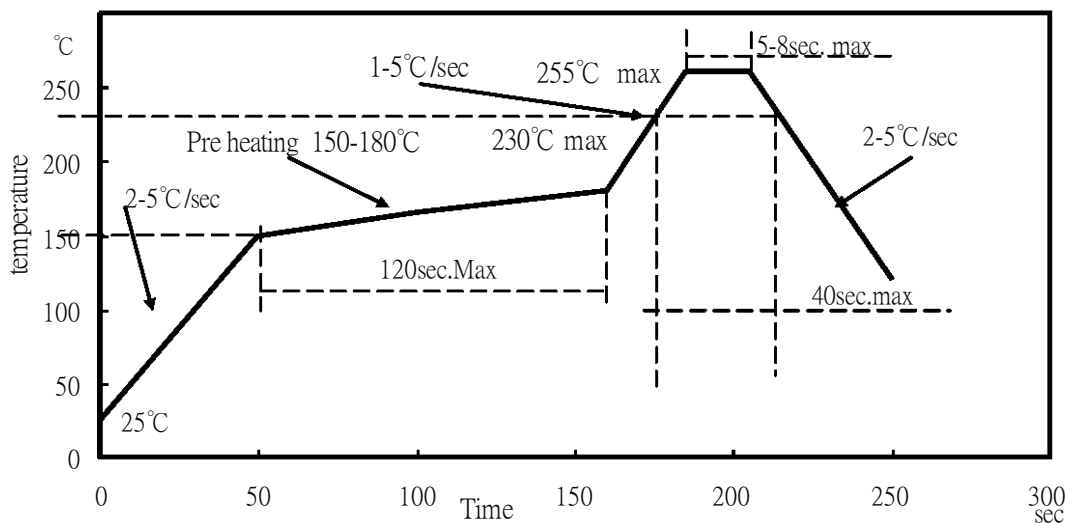
Fig.3 Typical Radiation Pattern 120°



Soldering Profile

- Recommended tin glue specifications: melting temperature in the range of 178~192 °C
- The recommended reflow soldering profile is as follows (temperatures indicated are as measured on the surface of the LED resin):

Lead Free solder



1. Reflow soldering should not be done more than two times. many times reflow soldering has destructive effect to the product.
2. When soldering, do not stress on the LEDs during heating.



Reliability test items and conditions:

NO	Item	Test Conditions	Time	Quantity	Ac/Re
1	Resistance to Soldering Heat	Tsld = 255°C, 10sec. 2 times	10 Sec	22 PCS	0/1
2	Temperature Cycle	-40°C~25°C~100°C~25°C 30min~5min~ 30min~5min	100 Cycles	22 PCS	0/1
3	Thermal Shock	H : +100°C 15min └ L : -40°C 15min	100 Cycles	22 PCS	0/1
4	High Temperature Storage	TEMP : 100°C	1000 Hrs	22 PCS	0/1
5	Low Temperature Storage	TEMP : -40°C	1000 Hrs	22 PCS	0/1
6	DC Operating Life	TEMP : 25°C	1000 Hrs	22 PCS	0/1
		If =12mA			
7	High Temperature / High Humidity	60°C / 90% RH	1000 Hrs	22 PCS	0/1

The technical information shown in the data sheets are limited to the typical characteristics and circuit examples of the referenced products. It does not constitute the warranting of industrial property nor the granting of any license.

Criteria for Judging the Damage:

Item	Symbol	Test Condition	Limit	
			Min	Max
Luminous Intensity	IV	DC=5V, Typical current	Init. Value*0.7	---
Resistance to Soldering Heat	---	DC=5V, Typical current	No NG or obvious damage	



■ Instructions of Soldering

1. Hand Solder should not be using hand soldering iron, suggest using heating operation, the use of solder paste melting point under 220 °C, the heating temperature set less than or equal to 255 °C, the time is not more than 5s, a welding is complete, multiple welding products are destructive
2. Handle the component along the side surface by using forceps or appropriate tools; do not directly touch or Handle the silicone lens surface, it may damage the internal circuitry.



■ Cautions

Storage condition

1. The operation of Temperatures and RH are : $\leq 30^{\circ}\text{C}$, RH<60%.
2. Once the package is opened, the products should be used within 8hours, Otherwise, more than the dehumidification process, dehumidifying conditions: reel 70 °C / 6 ~ 8 h, bulk LED150 °C / 3 h
3. Considering the operation environment of temperature and humidity ,we suggest our customers to use our products within three months (from production date), More than 3 months with toasted constant temperature 70 °C $\pm$ 5 °C for 6 ~ 8 hours desiccant (need to get rid of aluminum foil bag packing).
4. The outer diameter of the SMD pickup nozzle should not exceed the size of the LED to prevent air leaks. The inner diameter of the nozzle should be as large as possible. A pliable material is suggested for the nozzle tip to avoid scratching or damaging the LED surface during pickup. The dimensions of the component must be accurately programmed in the pick-and-place machine to insure precise pickup and avoid damage during production.

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5. LED operating environment and sulfur element composition cannot be over 100PPM in the LED contact usage material.
6. When external glue needed for LED application products, please make sure that the external glue matches the LED packaging glue. As most of LED packaging glue is silica gel, and it has strong Oxygen permeability as well as strong moisture permeability; in order to prevent external material from getting into the inside of LED, which may cause the malfunction of LED, the single content of Bromine element is required to be less than 900PPM, the single content of Chlorine element is required to be less than 900PPM, the total content of Bromine element and Chlorine element in the external glue of the application products is required to be less than 1500PPM
7. Please note that the electrostatic protection during processing, All devices, equipment and machinery must be properly grounded.

Products use manual

1. LED unfavorable use by constant voltage or more LED directly to the parallel use, advising clients to use in constant current mode.
2. When using LED products, please note that according to the grades of the same (voltage, brightness, color temperature) code together, different grades and different batches should be confirmed to meet customers' requirements to mix stick, lest produce color difference or brightness difference.

Part Numbering Code

S	3535	3	RGB	SK6812MINI	D0	01
SMD Type	Lead frame size	Chip qty	Color	Pd(watt)/nm/CCT/IC	Chip Code	Version